

Synthetic features: model importance and known roles

| Feature           | Description        | Role              | Control | Case  | Mean  SHAP |
|-------------------|--------------------|-------------------|---------|-------|------------|
| twist_signal_05   | Synthetic signal 5 | engineered signal | 27.5    | 80.0  | 0.09       |
| corona2_signal_03 | Synthetic signal 3 | engineered signal | 27.5    | 70.0  | 0.06       |
| corona2_signal_06 | Synthetic signal 6 | engineered signal | 75.0    | 31.25 | 0.06       |
| agilent_signal_01 | Synthetic signal 1 | engineered signal | 35.0    | 72.5  | 0.05       |
| twist_signal_02   | Synthetic signal 2 | engineered signal | 72.5    | 32.5  | 0.05       |
| agilent_signal_04 | Synthetic signal 4 | engineered signal | 68.75   | 31.25 | 0.05       |
| agilent_noise_024 | Noise peptide 24   | independent noise | 31.25   | 10.0  | 0.02       |
| agilent_noise_006 | Noise peptide 6    | independent noise | 30.0    | 52.5  | 0.02       |
| BMI               |                    |                   | 24.98   | 25.22 | 0.01       |
| Age               |                    |                   | 50.52   | 50.85 | 0.01       |